

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

CONSTRAINT-BASED PROBLEM SOLVING

QUESTION 1: HOSPITAL DOCTOR SHIFT SCHEDULING USING BACKTRACKING SEARCH

Problem Statement:

A hospital needs to assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night). The assignment must satisfy all given constraints. The objective is to use Backtracking Search to find a valid schedule.

Variables

Variable	Description
D1	Doctor 1
D2	Doctor 2
D3	Doctor 3

Domain

Doctor	Possible Shifts
D1	Morning, Afternoon
D2	Morning, Afternoon, Night
D3	Afternoon, Night

Constraints

Constraint No	Constraint
C1	D1 cannot work Night shift
C2	D2 must work before D3

C3	Only one doctor per shift
C4	D3 cannot work Morning shift
C5	All doctors must be assigned exactly one shift

Initial State

Doctor	Shift
D1	Unassigned
D2	Unassigned
D3	Unassigned

Step 1

Assign D1 → Morning

Doctor	Shift
D1	Morning
D2	Unassigned
D3	Unassigned

Constraint Status:

Constraint	Status
C1	Satisfied
C2	Not Checked
C3	Satisfied
C4	Not Checked
C5	Not Satisfied Yet

Step 2

Assign D2 → Afternoon

Doctor	Shift
D1	Morning

D2	Afternoon
D3	Unassigned

Constraint Status:

Constraint	Status
C1	Satisfied
C2	Pending
C3	Satisfied
C4	Not Checked
C5	Not Satisfied Yet

Step 3

Assign D3 → Night

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Final Constraint Verification

Constraint	Verification
D1 ≠ Night	✓ Satisfied
D2 before D3	✓ Afternoon before Night
One Doctor per Shift	✓ Satisfied
D3 ≠ Morning	✓ Satisfied
All Doctors Assigned	✓ Satisfied

Final Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

QUESTION 2: ROBOT NAVIGATION IN A GRID ENVIRONMENT USING BFS SEARCH

Problem Statement

A robot operates in a 5×5 grid environment. The robot starts from the Start position (S) and must reach the Goal position (G). Some cells contain obstacles that block movement. The robot can move only in four directions: Up, Down, Left, and Right. The objective is to find the shortest path from Start to Goal using Breadth First Search (BFS).

Environment Representation

Symbol	Meaning
S	Start Position
G	Goal Position
X	Obstacle
O	Open Cell

Sample Grid

S	O	O	O	O
O	X	X	O	O
O	O	O	X	O
O	X	O	O	O
O	O	O	O	G

Constraints

Constraint No	Constraint
C1	Movement allowed only Up, Down, Left, Right
C2	Cost of each move = 1
C3	Obstacles cannot be crossed
C4	Manhattan Distance heuristic used
C5	Optimal path must be found

Initial State

Item	Value
Start Node	S (0,0)
Goal Node	G (4,4)
Initial Queue	[(0,0)]

BFS Node Expansion Table

Step	Node Expanded	New Nodes Added to Queue
1	(0,0)	(0,1), (1,0)
2	(0,1)	(0,2)
3	(1,0)	(2,0)
4	(0,2)	(0,3)
5	(2,0)	(2,1), (3,0)
6	(0,3)	(0,4), (1,3)
7	(2,1)	(2,2)
8	(3,0)	(4,0)
9	(0,4)	(1,4)
10	(1,3)	(2,3) Blocked
11	(2,2)	(3,2)
12	(4,0)	(4,1)

13	(1,4)	(2,4)
14	(3,2)	(3,3)
15	(4,1)	(4,2)
16	(2,4)	(3,4)
17	(3,3)	(4,3)
18	(3,4)	(4,4) Goal Found

Queue Updates

Step	Queue Contents
Initial	[(0,0)]
After Step 1	[(0,1),(1,0)]
After Step 2	[(1,0),(0,2)]
After Step 3	[(0,2),(2,0)]
After Step 4	[(2,0),(0,3)]
...	...
Final	Goal Reached

Manhattan Distance Formula

Manhattan Distance =

$$|x_1 - x_2| + |y_1 - y_2|$$

Example

Current Node	Goal Node	Manhattan Distance
(0,0)	(4,4)	8
(2,2)	(4,4)	4
(3,4)	(4,4)	1
(4,4)	(4,4)	0

Optimal Path Found

Step	Position
1	(0,0)
2	(0,1)
3	(0,2)
4	(0,3)
5	(0,4)
6	(1,4)
7	(2,4)
8	(3,4)
9	(4,4)

Path Cost Calculation

Move	Cost
8 Moves	8×1

Total Cost = 8

Result

Item	Value
Start Node	(0,0)
Goal Node	(4,4)
Optimal Path	Found
Total Cost	8

Conclusion

Breadth First Search successfully explored the grid level by level and found the shortest path from the Start node to the Goal node while avoiding obstacles. The optimal path cost is 8, satisfying all given constraints.

QUESTION 3: AUTONOMOUS RESCUE ROBOT IN A DISASTER ENVIRONMENT

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The environment contains obstacles, blocked paths, and risky areas. The robot can observe only adjacent cells and must continuously update its knowledge while navigating. The objective is to reach the survivor safely while minimizing travel cost and risk using an appropriate AI search strategy.

Environment Components

Component	Description
S	Start Position
G	Survivor (Goal)
O	Safe Area
X	Obstacle
R	Risk Zone

Constraints

Constraint No	Constraint
C1	Robot can move Up, Down, Left, Right
C2	Obstacles cannot be crossed
C3	Robot observes only adjacent cells
C4	Each move costs 1 unit
C5	Risk zones add 2 extra cost units
C6	Total cost must be minimized
C7	Environment knowledge must be updated dynamically

Environment Representation

S	O	R	O	O
O	X	X	O	O
O	O	R	X	O
O	X	O	O	O
O	O	O	O	G

Initial State

Item	Value
Start Position	S
Goal Position	G
Initial Cost	0
Knowledge of Environment	Partial

Selected Search Strategy

Strategy	Reason
Uniform Cost Search (UCS)	Finds the least-cost path considering movement and risk costs

Step 1

Robot starts at S.

Position	Cost
S	0

Adjacent cells are observed.

Step 2

Robot evaluates neighboring cells.

Possible Move	Cost
Safe Cell (O)	1

Risk Zone (R)	3
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Robot chooses the lower-cost path.

Step 3

Robot updates environment knowledge.

Observation	Action
Obstacle Found	Avoid Path
Safe Path Found	Continue Search
Risk Zone Found	Compare Costs

Step 4

Uniform Cost Search expands the node with minimum cumulative cost.

Node	Total Cost
Safe Path	Lower
Risk Path	Higher

Lowest-cost node is selected for expansion.

Step 5

Robot continues exploration until Goal is reached.

Status	Result
Goal Reached	Yes
Survivor Located	Yes

Cost Calculation

Action	Cost
Normal Move	1
Risk Zone Entry	$1 + 2 = 3$

Example

Path Type	Total Cost
Safe Route	8
Risk Route	12

Robot selects the safe route because it has the minimum cost.

Application of AI Concepts

AI Concept	Application
Intelligent Agent	Robot perceives and acts in the environment
Rationality	Chooses the least-cost path
Percepts	Adjacent cell observations
Actions	Move Up, Down, Left, Right
Environment	Partially Observable and Dynamic
Search Strategy	Uniform Cost Search

Final Path

Step	Position
1	S
2	O
3	O
4	O
5	O
6	O
7	O
8	O
9	G

Result

Item	Value
Search Strategy Used	Uniform Cost Search
Goal Reached	Yes
Survivor Found	Yes
Minimum Cost Path	Obtained

Conclusion

Uniform Cost Search successfully guides the rescue robot through a partially observable disaster environment. The robot continuously updates its knowledge, avoids obstacles, minimizes risk, and reaches the survivor using the least-cost path. This approach ensures efficient and intelligent decision-making in rescue operations.